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AI-Based Shoreline Mapping from Satellite Imagery Using Binary Semantic Segmentation: Insights from Mousuni Island.

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Shoreline retreat is a major issue of the coastal resilience of monsoon-based marine setting, with the process of shoreline development being arbitrated by the intricate kinship between the weather climate and the processes of sediment transportation and the coastal geomorphology. The western Indian Sundarbans especially the Mousuni Island is especially at risk because of the high tidal currents, interactions of waves and tides, and reduced sediment supply leading to increased hydro-dynamic stress and eventual shoreline instability. This paper introduces a 35-year (1990-2025) spatiotemporal model of shoreline change on the basis of an interdisciplinary geospatial and deep learning platform through multi-temporal Landsat data. U-Net and DeepLabV3+ are two state-of-the-art semantic segmentation networks that were implemented and compared systematically regarding automated delineation of the shoreline. Intersection over Union (IoU), Dice coefficient, pixel accuracy, and composite loss measures were used as quantitative measures of model performance. The findings show that DeepLabV3+ had better performance on all evaluation parameters with a mean IoU of 0.93, mean Dice coefficient of 0.96, and mean pixel accuracy of 0.96 and a lower mean loss of 0.15 than U-Net which had a mean IoU of 0.85, a mean Dice coefficient of 0.92 and a lower mean pixel accuracy of 0.96 and a higher mean loss of 0.23. This can be explained by the fact that the high performance of DeepLabV3+ is explained by its higher capability of extracting a multi-scale contextual feature and a better ability to delineate boundaries that are able to identify robust shoreline detection in heterogeneous coastal environments. According to the long-term analysis, dominant erosional patterns and emerging and growing coastal vulnerability can be outlined, and with ongoing adoption of rising AI-controlled geospatial frameworks, climate-resistant coastal monitoring and sustainable shoreline management in deltaic settings are highly valued.

Keywords: Shoreline Extraction, Semantic segmentation, Unet, DeeplabV3+, Remote Sensing.